Module-8

Topic-1:Transfer learning:What, Why, and When?

Acknowledgement

Thanks to the following courses and corresponding researchers for making their teaching/research material online

- Mike X Cohen's Master Deep Learning course (most contents taken from this Udemy Course)
- AI Tools such as Chat GPT, Grok, etc
- Introduction to Deep Learning, University of Illinois at Urbana-Champaign
- Introduction to Deep Learning, Carnegie Mellon University
- Convolutional Neural Networks for Visual Recognition, Stanford University
- Natural Language Processing with Deep Learning, Stanford University
- And Many More .....

Topic -1

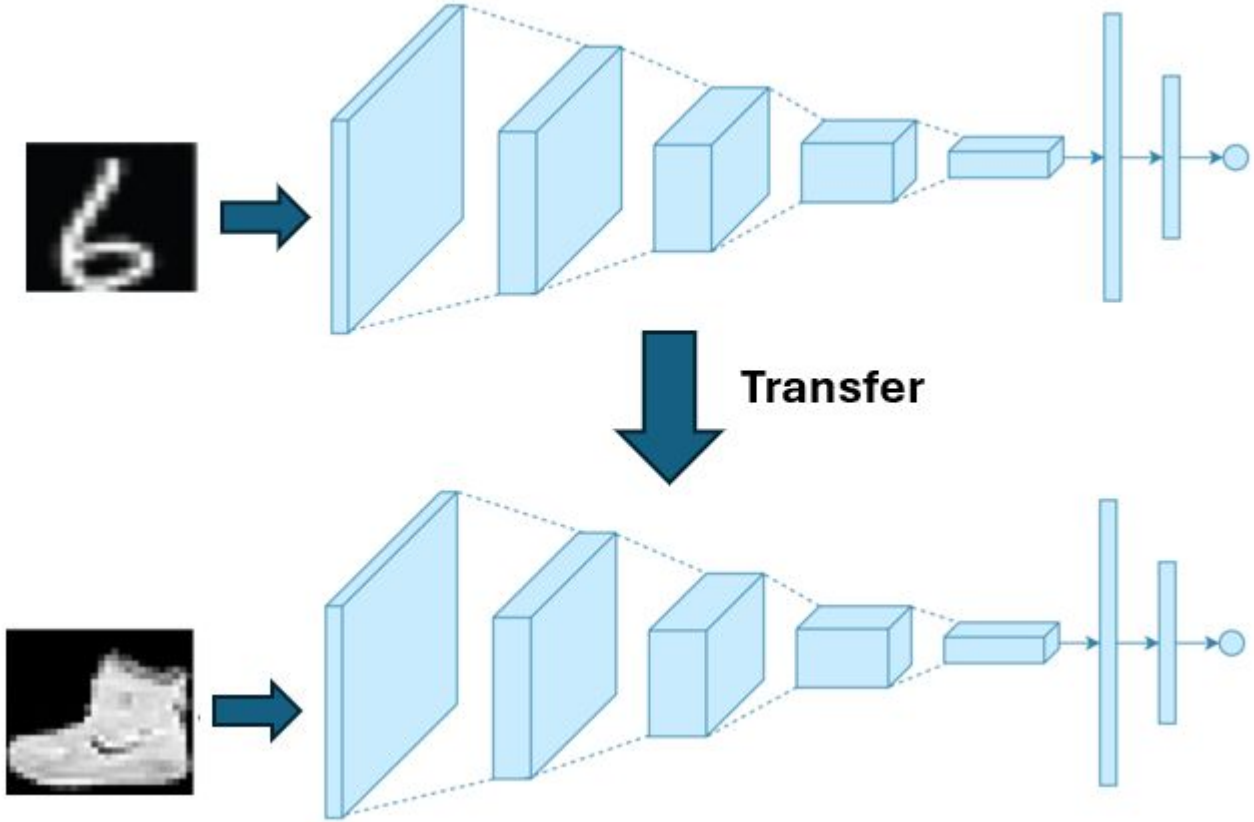
Transfer learning:What, Why, and When?

Transfer Learning; What, Why, and when?

In this topic, you will learn

- What “transfer learning” means.
- Why transfer learning is useful in DL.
- When to use transfer learning
- How to use transfer learning

What is transfer learning?



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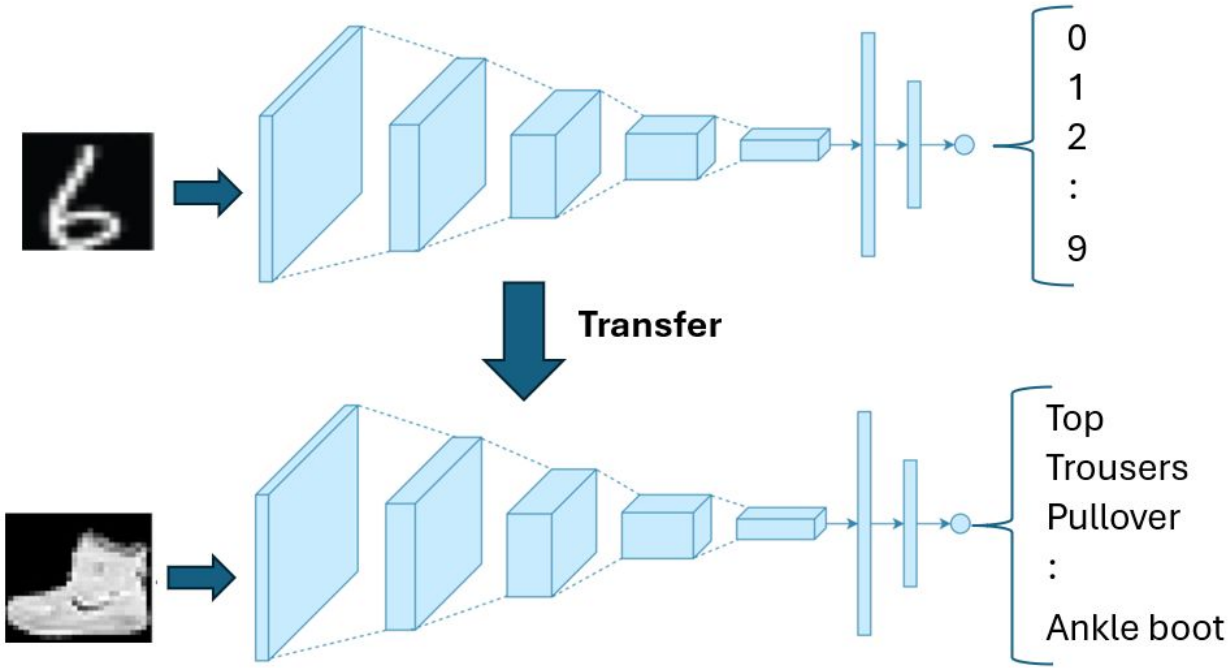

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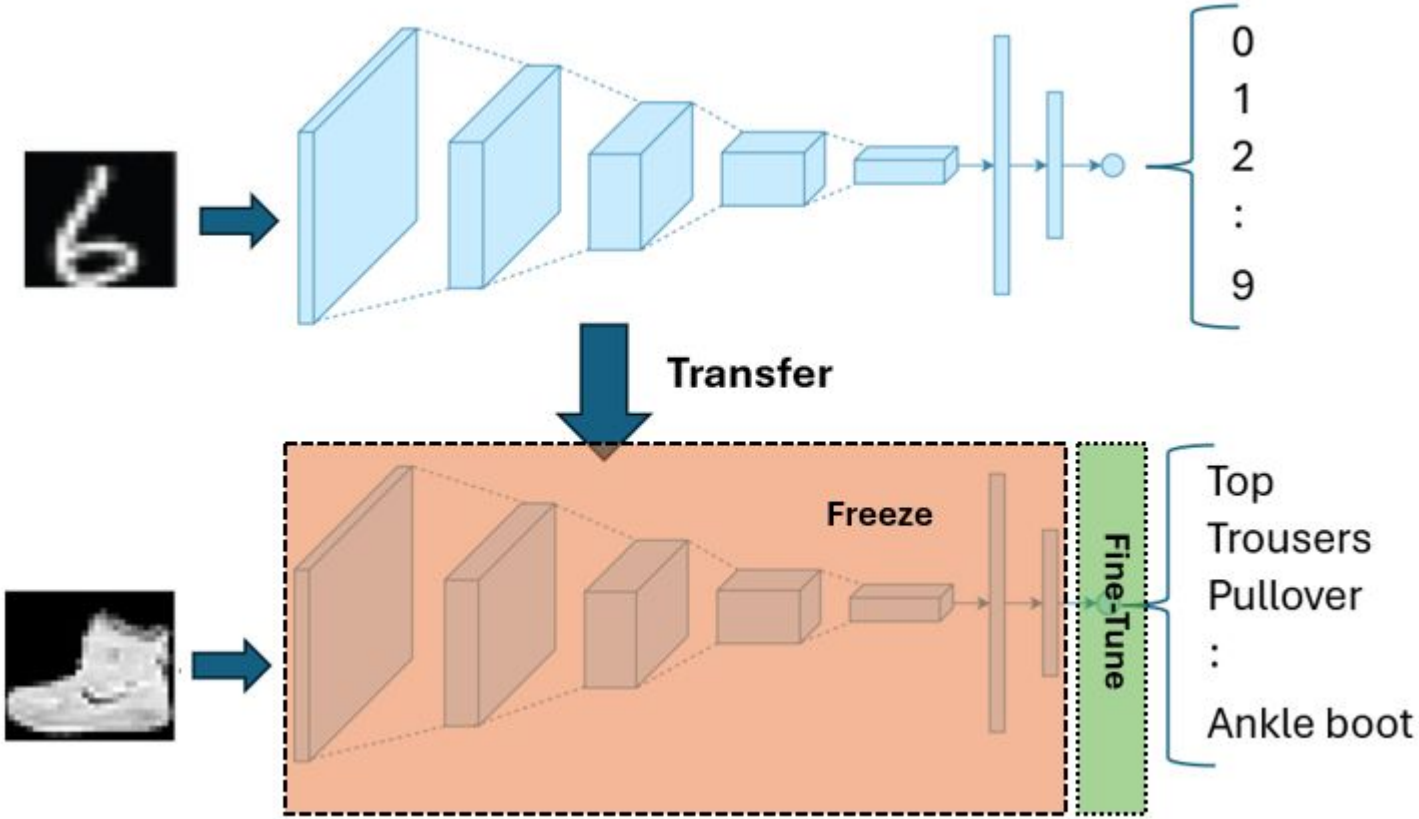

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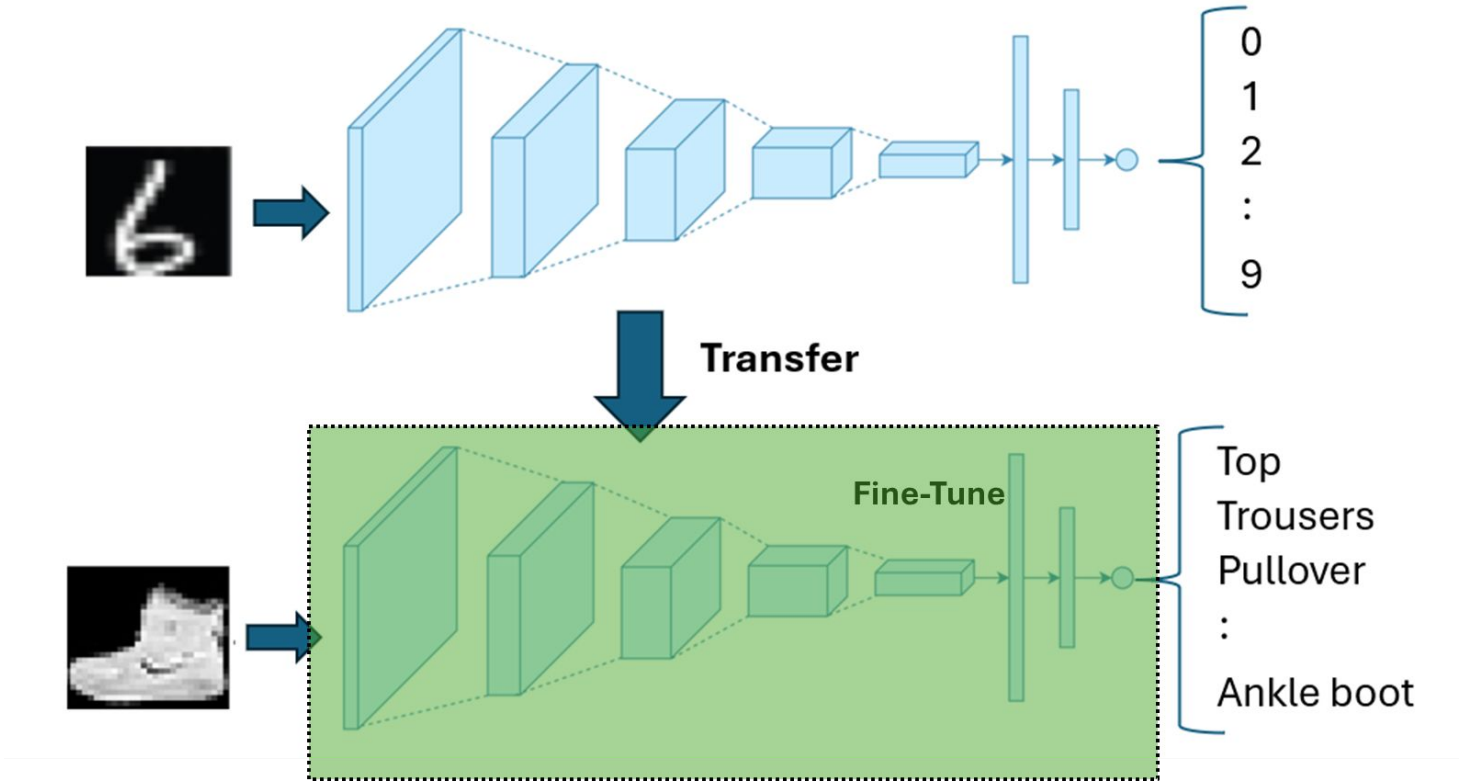

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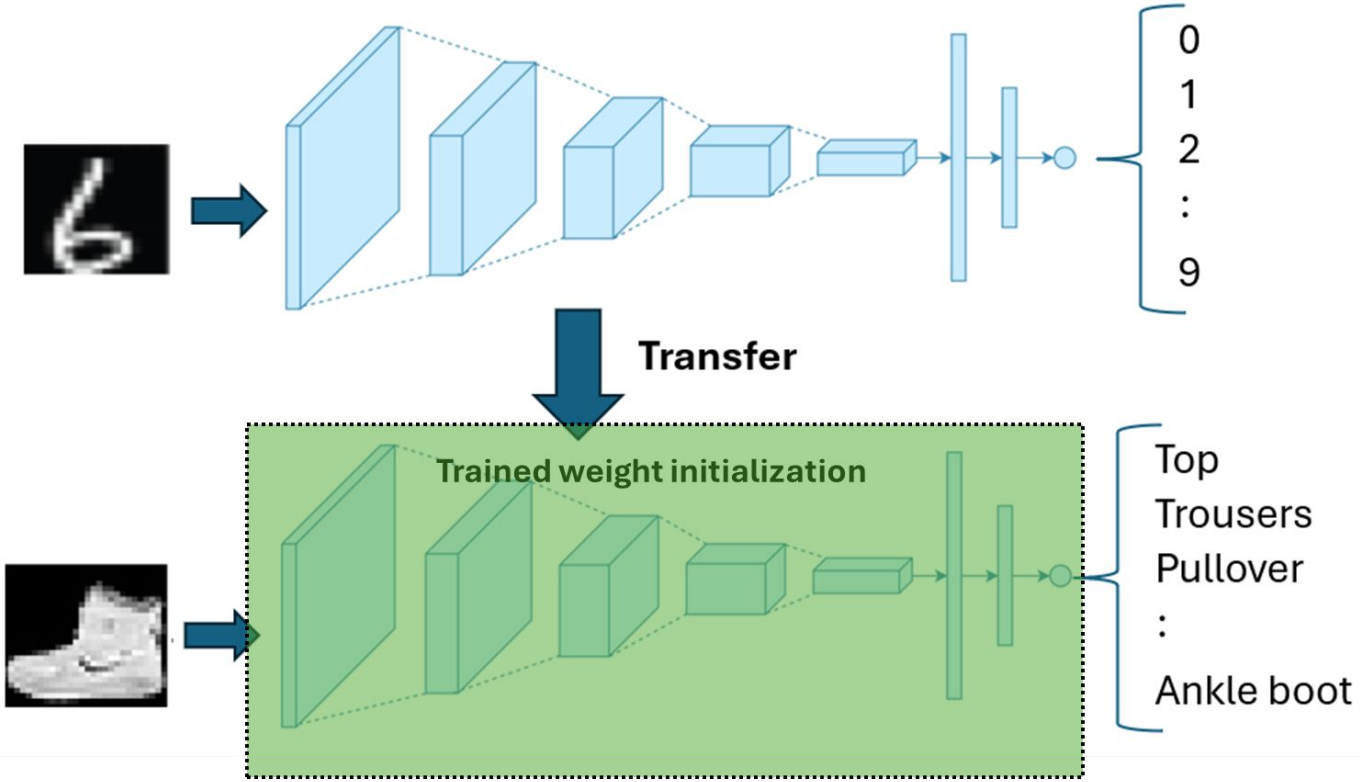

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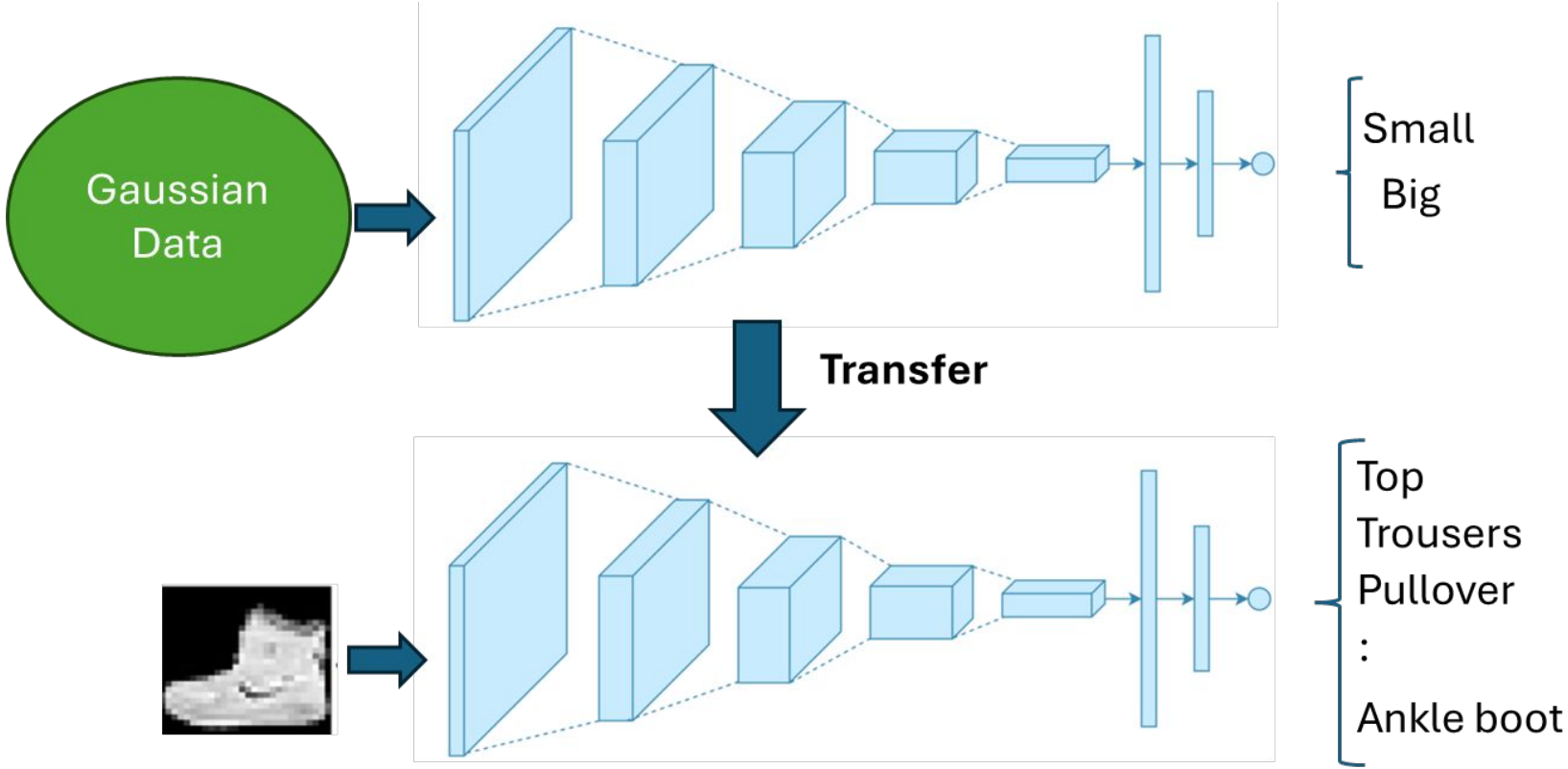

Why is transfer learning useful?

Transfer learning allows you to

- Leverage HUGE datasets and existing (source) models to solve new problems.
- Fine-tune existing trained models to your unique data.
- Deploy models quickly and with limited computational and data resources.

When is transfer learning appropriate?

Use transfer learning when

- Your problem is similar to a problem that someone else has solved (and published their model)
- The model was trained on a lot more data than what you have.
- The model is deep;shallow models are less likely to transfer well unless the target data are very similar.

Is transfer learning appropriate here?

- Fine-tune AlexNet to distinguish different models of BMWs(right)
- Fine-tune AlexNet to predict credit card fraud(wrong)
- Train a model on US housing sales data and then fine-tune that model to India housing sales data(don't' know).

How to use transfer learning

- Decide whether it is appropriate:(1) Do you have insufficient complex data and (2) does a well-trained model exist?
- Decide which(if any) layers to freeze: The closer your data to the source model, the more layers you can freeze.
- Fine-tune the target model as much as possible.
- Inspect the model and results carefully; never assume that transfer will be successful.

Transfer Learning; What, Why, and when?

In this topic, you have learned about

- What “transfer learning” means.
- Why transfer learning is useful in DL.
- When to use transfer learning
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Topic -2

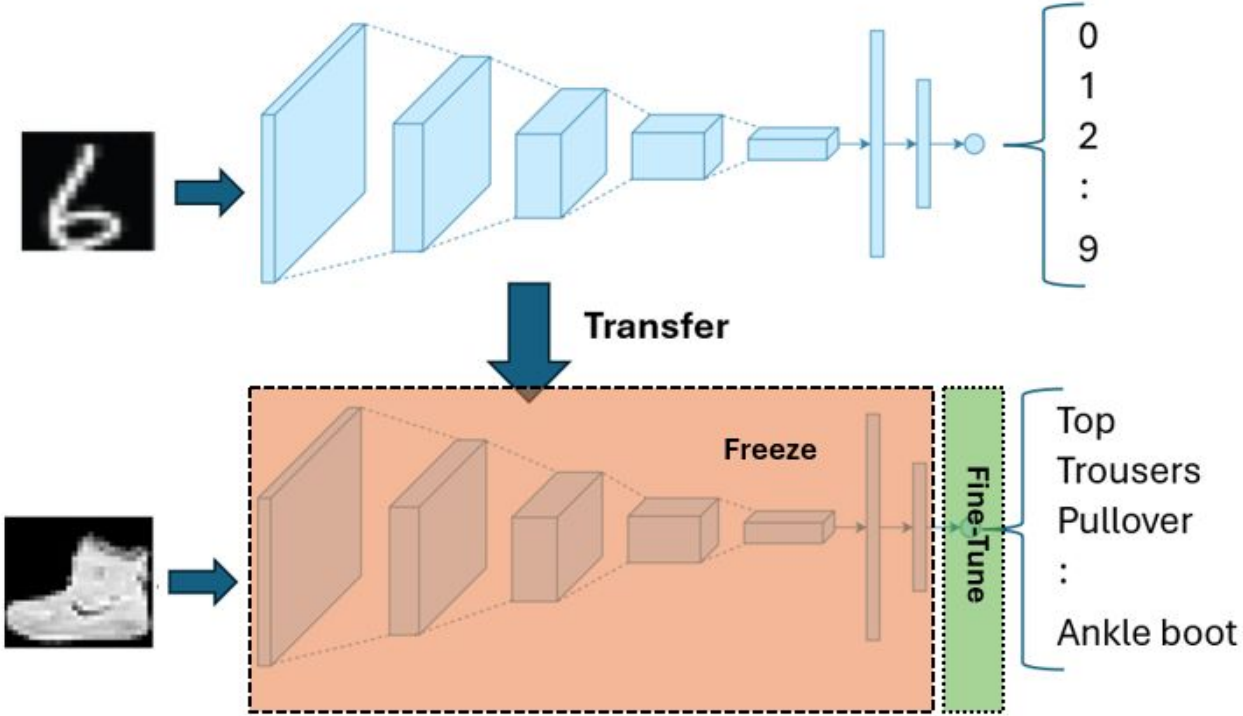
Transfer learning:MNIST->FMNIST

Transfer Learning; What, Why, and when?

In this topic, you will learn

- How to implement transfer learning in PyTorch.
- How to copy model parameters

What is transfer learning?



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Transfer Learning; What, Why, and when?

In this topic, you have learned about

- How to implement transfer learning in PyTorch.
- How to copy model parameters

Topic -3

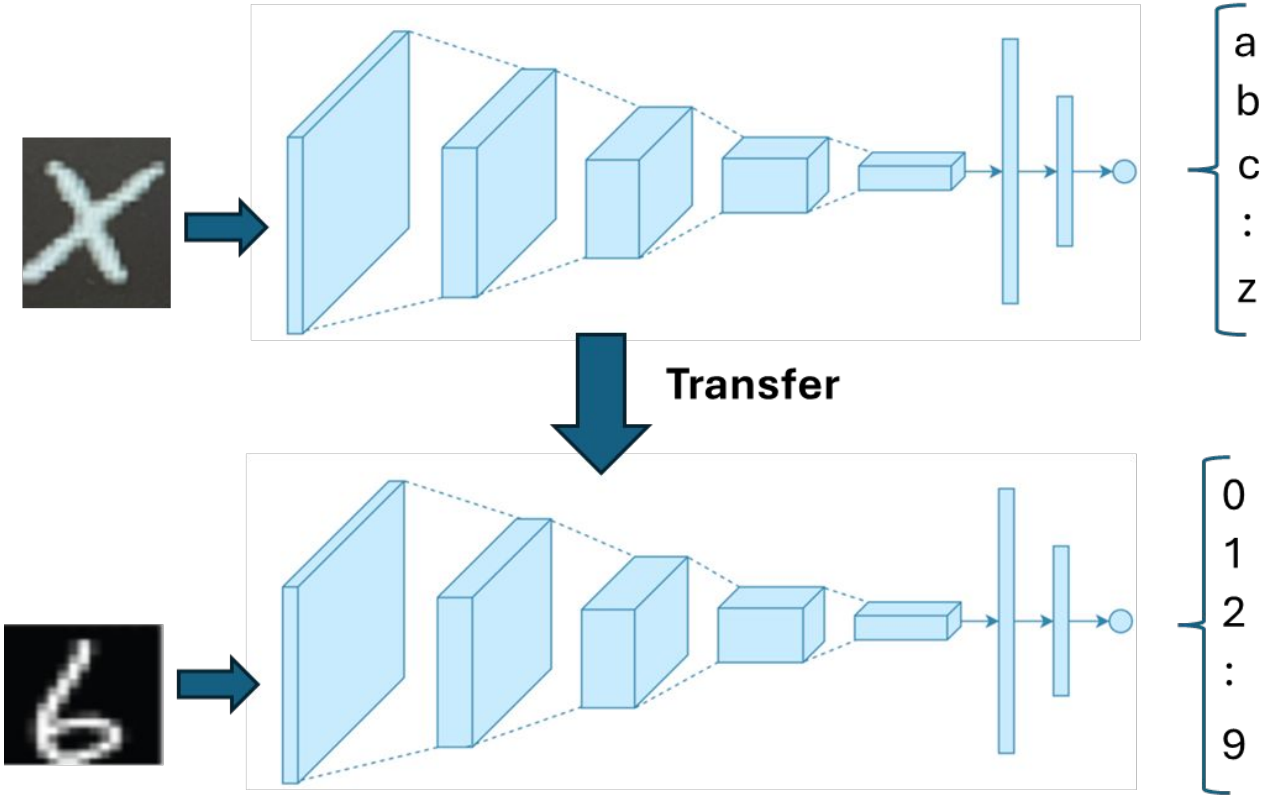
Transfer learning:Letters to numbers

Transfer Learning; What, Why, and when?

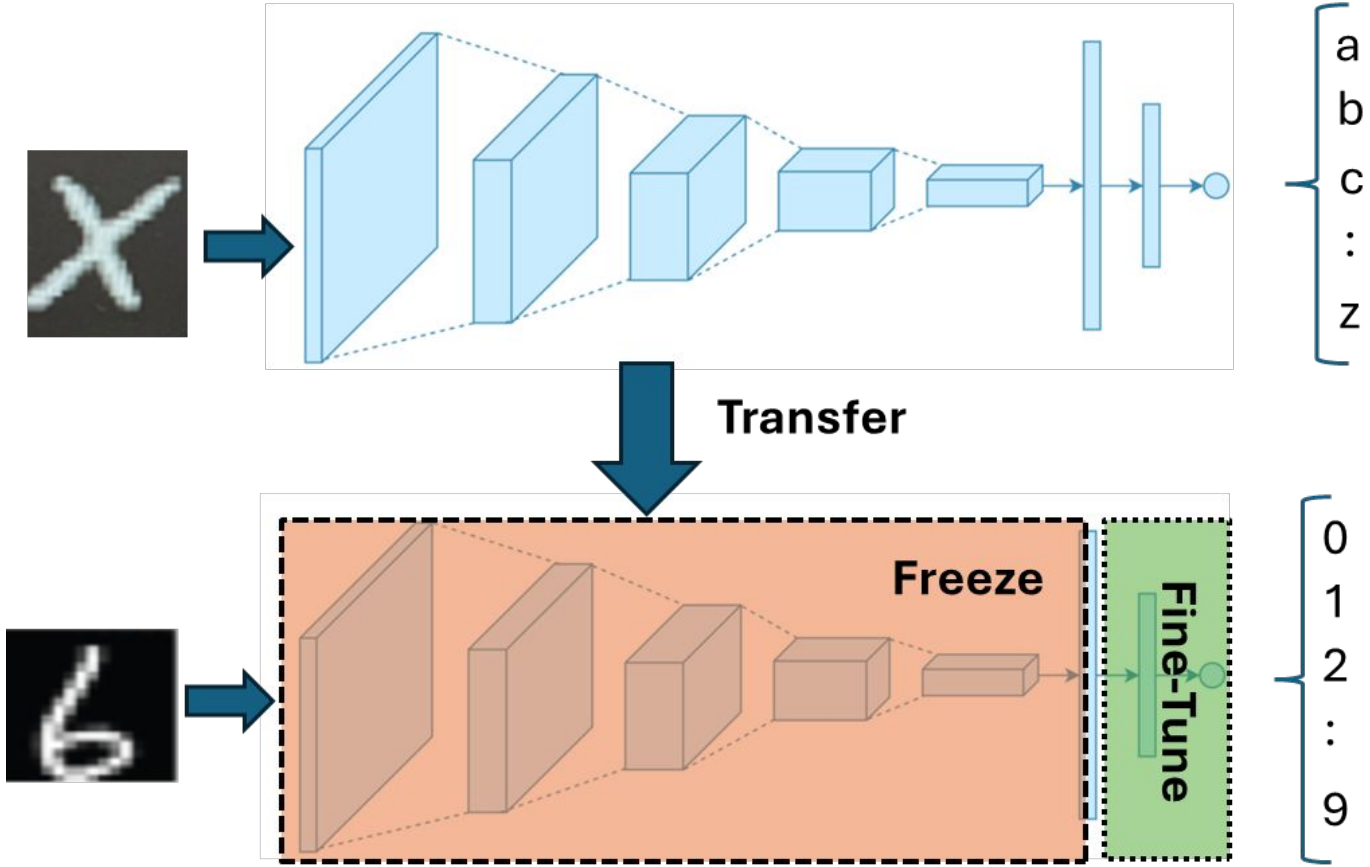
In this topic, you will learn

- Gain more experience working with transfer learning
- Learn how to freeze only the categorization layers in re-training

What is transfer learning ?



What is transfer learning ?



Our Objective

- Import both ‘letter’ and ‘digits’
- Train on letters for 5 epochs
- Transfer the model to digits
- Fine-tune (1) the entire model, and (2) only the categorization layers. Train only one epoch.

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Transfer Learning; What, Why, and when?

In this topic, you have explored how to

- Gain more experience working with transfer learning
- Learn how to freeze only the categorization layers in re-training

Topic -4

Transfer learning:famous CNN architectures

Transfer learning:famous CNN architectures

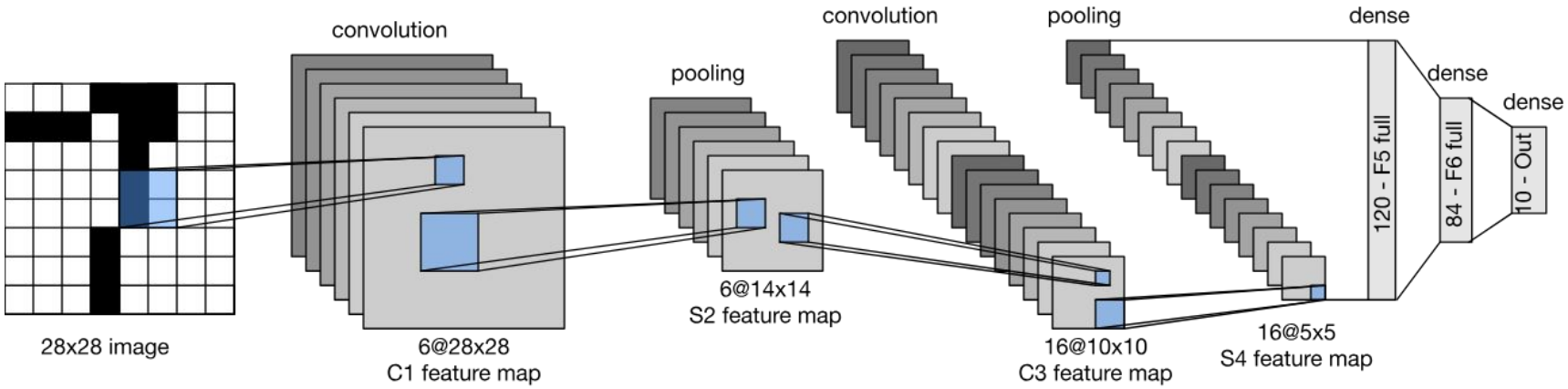
In this topic, you will explore

- Different architecture of CNN models

TABLE 1. State of art CNN models.

Model	Proposed Year	No of layers	No of Parameters
AlexNet	2012	8	62.3 M
DenseNet121	2016	121	6.99 M
EfficientNet_b0	2019	66	5.3 M
GoogLeNet	2014	22	5.64 M
MnasNet1_0	2018	39	4.4 M
MobileNet_v1	2017	28	3.25 M
Resnet 50	2015	50	23.59 M
ShuffleNet_v1	2018	27	5.4 M
VGG-16	2014	16	134.42 M
VGG-19	2014	19	143.67 M

LeNet


convolution
pooling
convolution
pooling
dense
dense
dense
28x28 image
6@28x28
C1 feature map
6@14x14
S2 feature map
16@10x10
C3 feature map
16@5x5
S4 feature map
120 - F5 full
84 - F6 full
10 - Out

AlexNet

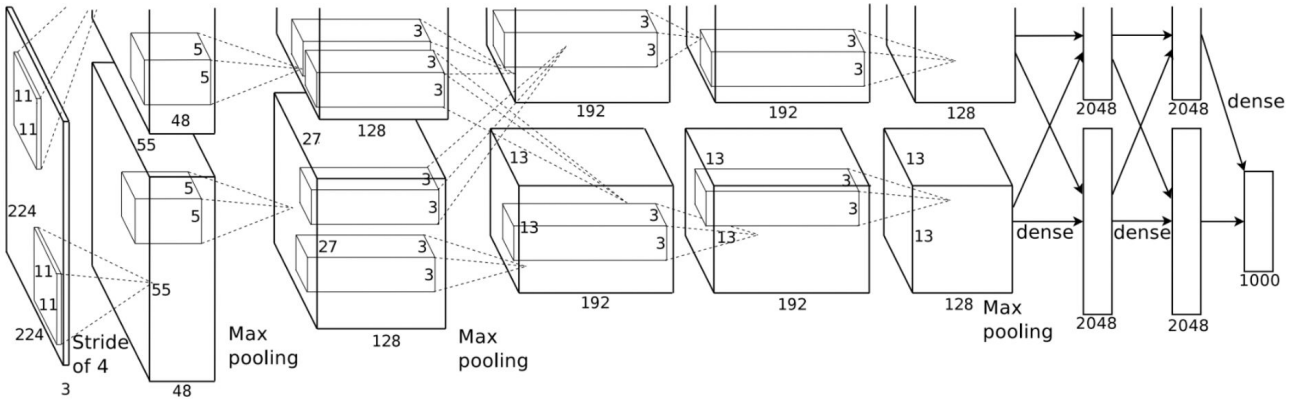
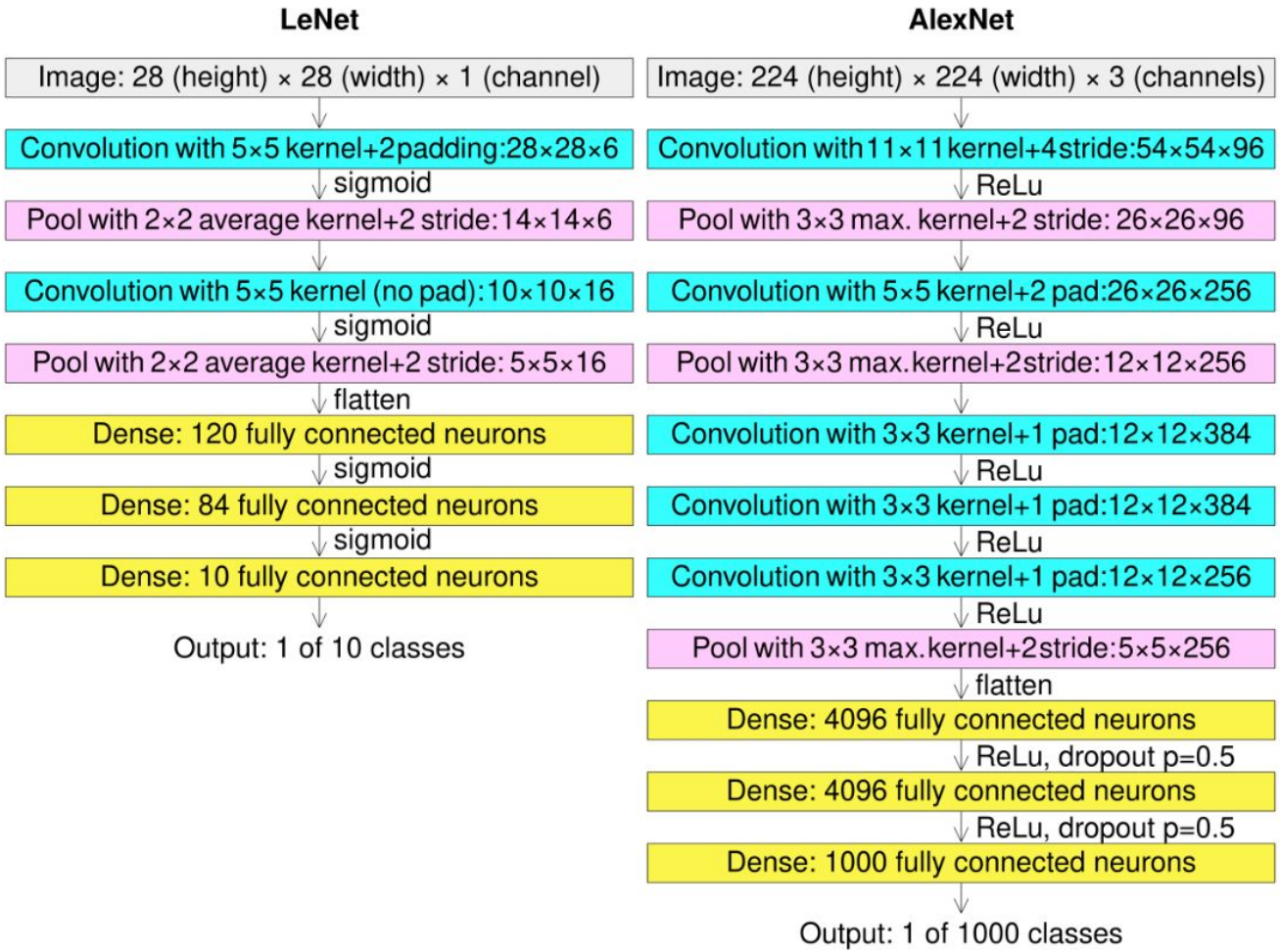
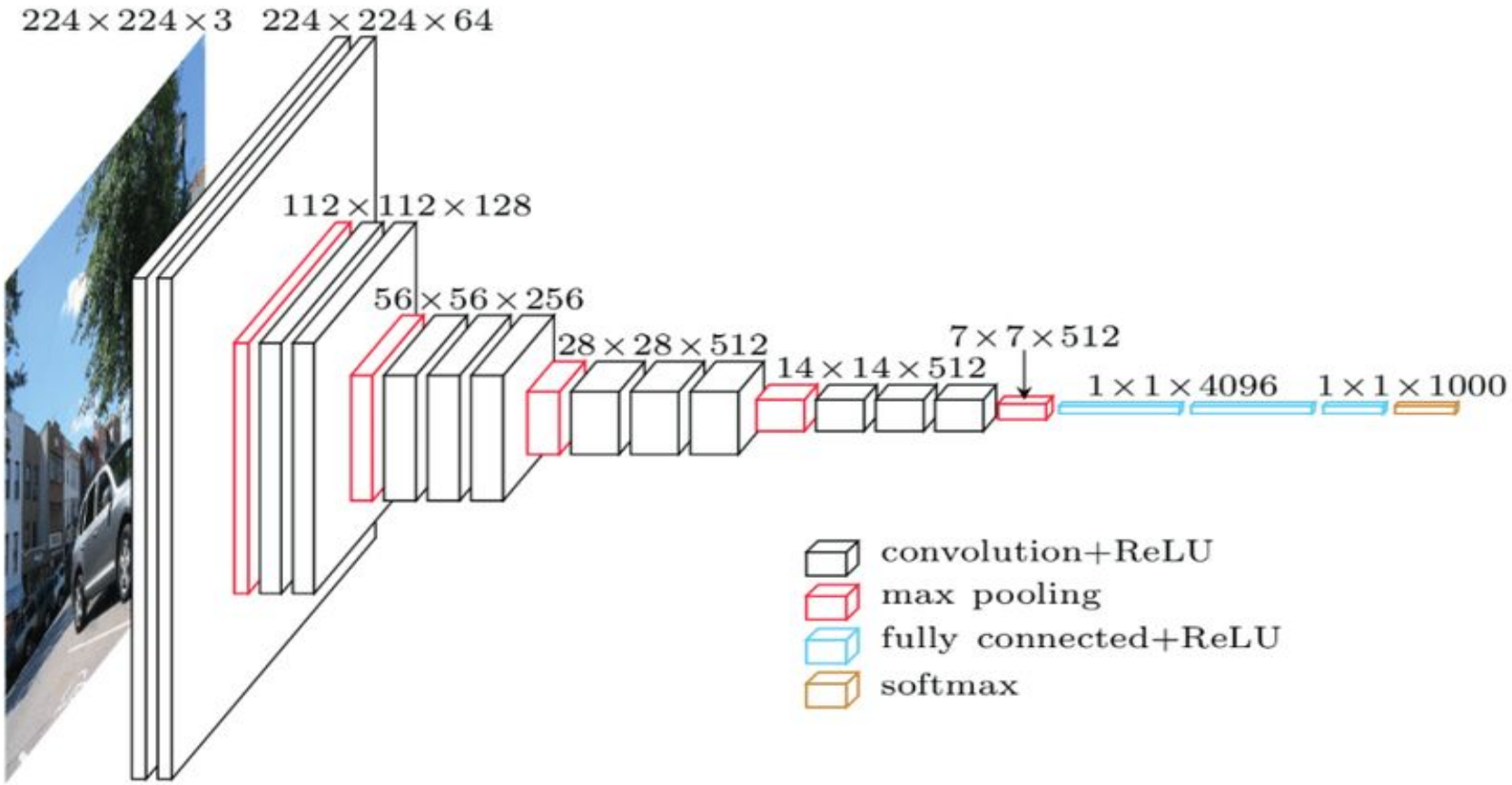
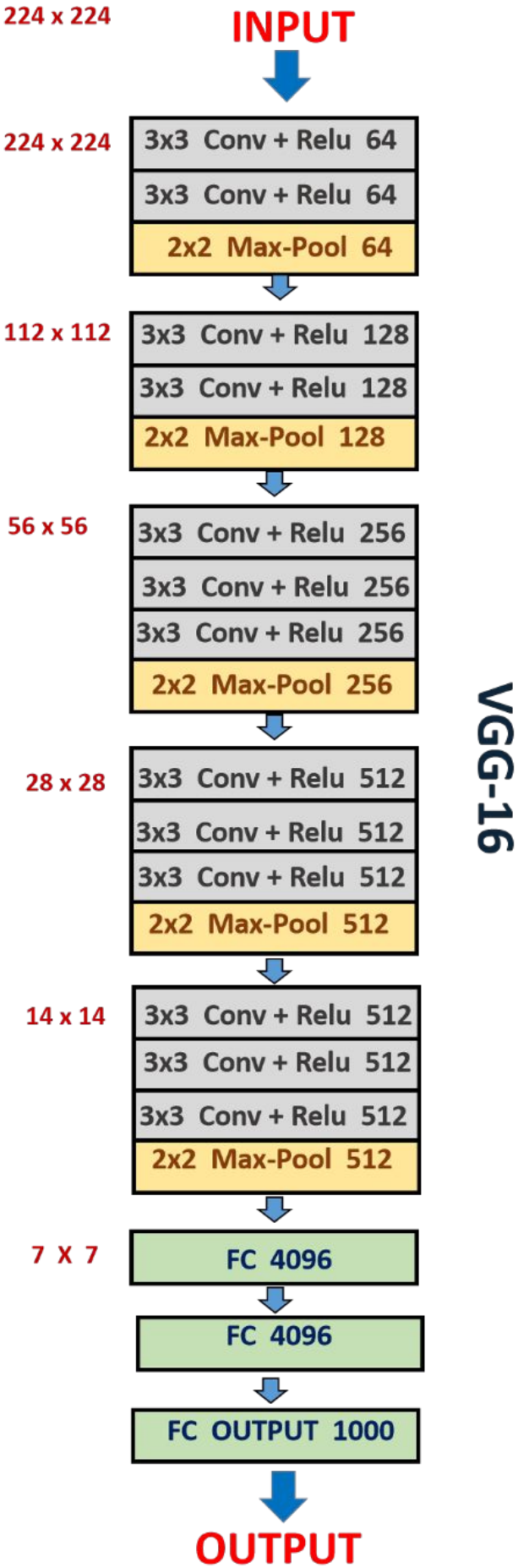


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

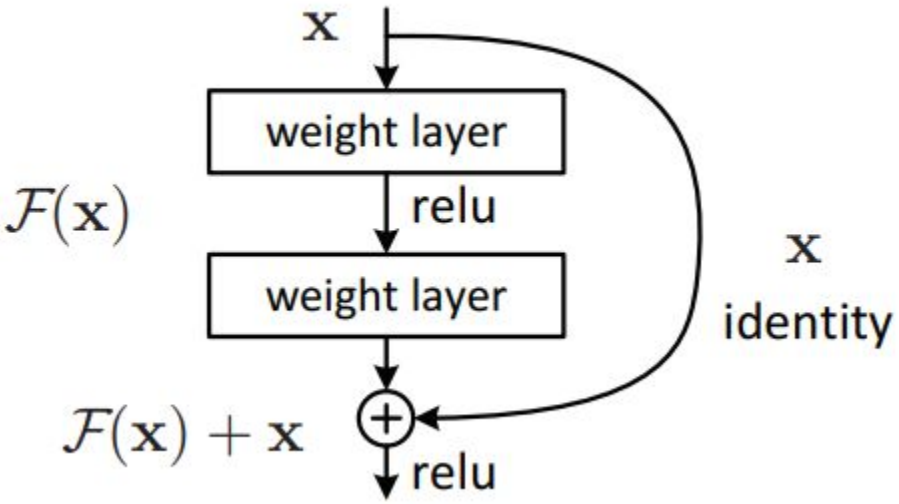

LeNet
Image: 28 (height) × 28 (width) × 1 (channel)
Convolution with 5×5 kernel+2padding:28×28×6
sigmoid
Pool with 2×2 average kernel+2 stride:14×14×6
Convolution with 5×5 kernel (no pad):10×10×16
sigmoid
Pool with 2×2 average kernel+2 stride: 5×5×16
flatten
Dense: 120 fully connected neurons
sigmoid
Dense: 84 fully connected neurons
sigmoid
Dense: 10 fully connected neurons
Output: 1 of 10 classes
AlexNet
Image: 224 (height) × 224 (width) × 3 (channels)
Convolution with 11×11 kernel+4 stride:54×54×96
ReLu
Pool with 3×3 max. kernel+2 stride: 26×26×96
Convolution with 5×5 kernel+2 pad:26×26×256
ReLu
Pool with 3×3 max.kernel+2stride:12×12×256
Convolution with 3×3 kernel+1 pad:12×12×384
ReLu
Convolution with 3×3 kernel+1 pad:12×12×384
ReLu
Convolution with 3×3 kernel+1 pad:12×12×256
ReLu
Pool with 3×3 max.kernel+2stride:5×5×256
flatten
Dense: 4096 fully connected neurons
ReLu, dropout p=0.5
Dense: 4096 fully connected neurons
ReLu, dropout p=0.5
Dense: 1000 fully connected neurons
Output: 1 of 1000 classes

VGGNet


224 × 224 × 3
224 × 224 × 64
112 × 112 × 128
56 × 56 × 256
28 × 28 × 512
14 × 14 × 512
7 × 7 × 512
1 × 1 × 4096
1 × 1 × 1000
convolution+ReLU
max pooling
fully connected+ReLU
softmax


224 x 224
INPUT
224 x 224
3x3 Conv + Relu 64
3x3 Conv + Relu 64
2x2 Max-Pool 64
112 x 112
3x3 Conv + Relu 128
3x3 Conv + Relu 128
2x2 Max-Pool 128
56 x 56
3x3 Conv + Relu 256
3x3 Conv + Relu 256
3x3 Conv + Relu 256
2x2 Max-Pool 256
28 x 28
3x3 Conv + Relu 512
3x3 Conv + Relu 512
3x3 Conv + Relu 512
2x2 Max-Pool 512
14 x 14
3x3 Conv + Relu 512
3x3 Conv + Relu 512
3x3 Conv + Relu 512
2x2 Max-Pool 512
7 X 7
FC 4096
FC 4096
FC OUTPUT 1000
OUTPUT
VGG-16

ResNet



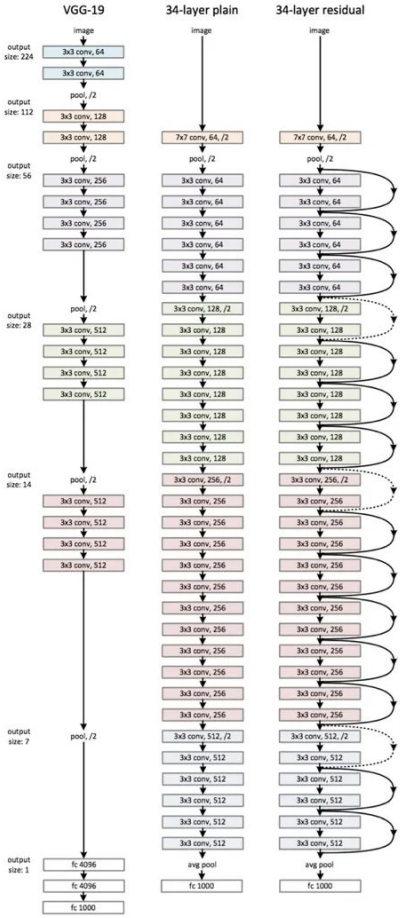


Figure 3. Example network architectures for ImageNet. **Left**: the VGG-19 model [41] (19.6 billion FLOPs) as a reference. **Middle**: a plain network with 34 parameter layers (3.6 billion FLOPs). **Right**: a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. **Table 1** shows more details and other variants.

Pytorch pre-trained model

The following classification models are available, with or without pre-trained weights:

- AlexNet
- ConvNeXt
- DenseNet
- EfficientNet
- EfficientNetV2
- GoogLeNet
- Inception V3
- MaxVit
- MNASNet
- MobileNet V2
- MobileNet V3
- RegNet
- ResNet
- ResNeXt
- ShuffleNet V2
- SqueezeNet
- SwinTransformer
- VGG
- VisionTransformer
- Wide ResNet

Transfer learning:famous CNN architectures

In this topic, you have explored

- Different architecture of CNN models

Topic -5

Transfer learning:with ResNet-18

Transfer learning:with ResNet-18

In this topic, you will explore

- ResNet-18 architecture of CNN models

ResNet architecture

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

ures for ImageNet. Building blocks are shown in brackets (see also Fig. 5), with the numbers of block

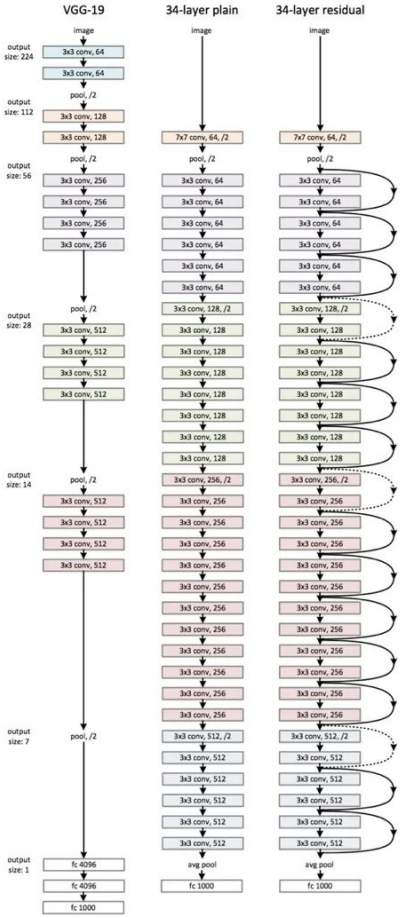
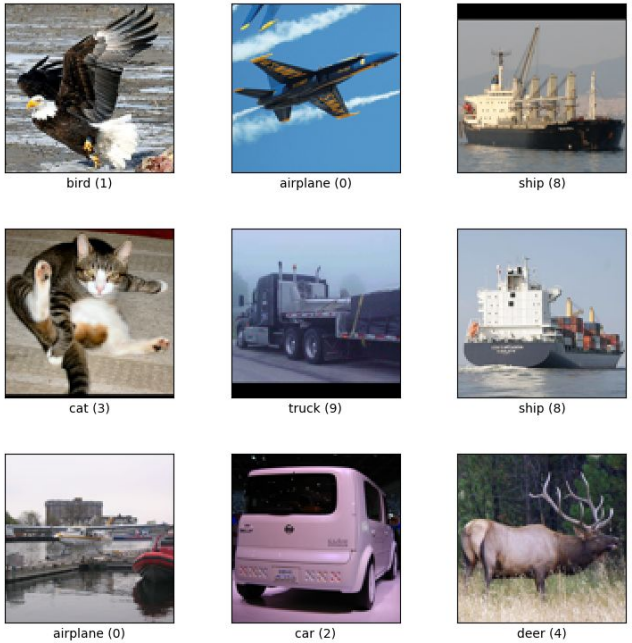


Figure 3. Example network architectures for ImageNet. **Left**: the VGG-19 model [41] (19.6 billion FLOPs) as a reference. **Middle**: a plain network with 34 parameter layers (3.6 billion FLOPs). **Right**: a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. **Table 1** shows more details and other variants.

STL10 datasets



The plan

1. Import the image data
2. Download a pretrained resnet18
3. Freeze the model
4. Replace the final layer
5. Fine-tune only the output layer

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Transfer learning:with ResNet-18

In this topic, you have explored

- Resnet-18 architecture of CNN models

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